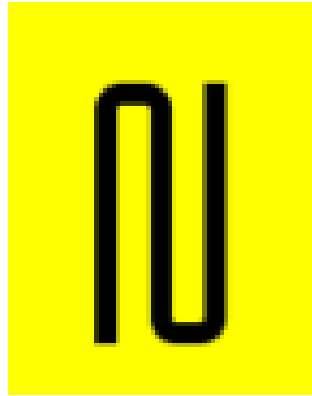




**MODULE 2 ASSIGNMENT**  
**BAN6430 DATA MODELLING AND MINING**  
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## 1 Part 1 – Identifying All Inconsistencies

Inconsistencies to be fixed:

1. File in txt to be worked on and saved to csv
2. Missing values in the columns
3. To check on duplicate values
4. To re-arrange the index sequence

## 2 Part 2 – Techniques to be Used

### 1. Data Collection

Upon downloading the txt file, I opened Jupyter Notebook to work on the file.

I import pandas, and then import dataset in txt, to be converted to csv.

Codes used as below:

```
import pandas as pd
input_txt_file = 'dataset.txt'
output_csv_file = 'dataset.csv'
df = pd.read_csv(input_txt_file, sep=',') #this is to separate data into columns in csv
df.head() #this is to check the resulting data frame header
```

### 2. Data Transformation

I convert the file to csv by using code below

```
df.to_csv(output_csv_file, index=False)
print(f'Data from '{input_txt_file}' saved to '{output_csv_file}' successfully.') #this is to
confirm successful conversion to csv.
df = pd.read_csv(input_txt_file, sep=',') #to read file and see the current data
df
```

### 3. Data Reduction 1

I checked for duplicates with below codes

```
df.drop_duplicates()
```

But apparently there's no duplication of data.

#### 4. Data Reduction 2

I take out all Nan values to ensure that data is valid for use by using code below:

```
df = df.dropna()
```

By taking out the Nan values, from 100 rows of data, there left only 68 rows of data.

#### 5. Data Normalization

I normalize the sequence of the data by using code below:

```
df=df.reset_index(drop=True)
```

### 3 Part 3 – Data saved into excel

The txt dataset had been worked on and cleaned and saved into csv excel file as requested.

### 4 Part 4 – Challenges & Opportunities

Challenges:

1. Data sample is too small
2. Purchase data is not detailed, only listing category of items whether it's shoes or clothes.
3. Advertising agency is not detailed
4. No contact details of the customers to seek feedback
5. No quantity of purchased items

Opportunities:

1. Company can use data to see purchase preference based on gender
2. Company can use data to see purchase preference based on age
3. Company can use data to see effectiveness of marketing each marketing channel
4. Company can use product rating to work on categories requiring improvement
5. Company can use this dataset to build a more detailed dataset for future reference and improvement.

## 5 References

1. Alex the Analyst. 2023. Data Cleaning in Pandas.  
[https://www.youtube.com/watch?v=bDhvCp3\\_IYw](https://www.youtube.com/watch?v=bDhvCp3_IYw)
2. My Github account <https://github.com/Kinta264/BAN6430-Module-2-Assignment>